Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/30/2009
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2006	**End Date:**	09/30/2011
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Flight definition NOTE: Extended to 9/30/2011, per NSSC info (9/2009) NOTE: NEW grant, resulting from grants reinstatement directive in 8/2006, per D. Stocker/GRC (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell (Yale University)		
Grant/Contract No.:	NNX07AB33A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum - namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform from which to understand better the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth and oxidation processes in soot formation models. The goals of this research will be the generation of modified kinetic mechanisms for hydrocarbon fuels that are able to model effectively diffusion flame structure under a larger parameter range than existing mechanisms, and the development of sub-models for soot formation that are capable of predicting both high and low soot loading levels in hydrocarbon flames of various fuels (specifically methane and ethylene).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place.
Task Progress:	Recent Progress With the successful completion of the Science Concept Review for our experiments on coflow laminar diffusion flames to be performed on the International Space Station (ISS), we have concentrated on further refinement and characterization of flame configurations and diagnostics that will allow us to take greatest advantage of the long periods of high-quality microgravity that are available on the ISS. This must be done in a way that will ensure compatibility with the Combustion Integrated Rack (CIR), which is now in place and running experiments on the ISS. A detailed examination of the flow fields and flames produced by a redesigned burner was undertaken. Velocity measurements were performed via a hot-wire anemometer in order to verify a top-hat profile exiting the burner coflow, ensure a fully developed flow exiting the fuel tube, and observe boundary-layer effects of differing burner configurations. Asymmetries in previous flame images have been investigated as well. Current work includes exploring other flame configurations that require a smaller amount of coflow air. We have also continued the measurement of sooting flames at normal gravity. We have worked with researchers at NASA Glenn on the specification of the measurement techniques to be used in the proposed microgravity measurements aboard the ISS. Techniques include color imaging, UV imaging of chemiluminescence, soot pyrometry for temperature and soot volume fraction, thin filament pyrometry for flame temperatures in nonsooting regions of the flame, and laser extinction measurements for soot volume fraction. Color photography will be used to monitor the appearance and stability of the flames at the various dilution levels. These images can be compared with the results of the computations. Using a relatively simple measurement technique of imaging line-of-sight flame emission using a color digital camera, it is possible to make a number of observations on the flame. Such things as flame shape and size, lift-off height, and qualitative distributions of CH* and soot can be observed. A paper describing techniques for making the comparison between experiments and computations as quantitative as possible, even with relatively simple diagnostic techniques, has been published. Microgravity Modeling The computational models used in our early flame studies included the chemical mechanisms of both GRI Mech 2.11 and a simple 26-species, C2 hydrocarbon mechanism developed at Yale. Both mechanisms clearly showed that buoyancy plays a role in both the size and shape of the laminar diffusion flame. Good comparisons between computations and experiments were obtained using the 26-species C2 mechanism for our standard, moderately diluted flame consisting of 65% methane and 35% nitrogen. However, the agreement in lift-off heights deteriorated rapidly as the dilution level in the flame was increased. Analysis of this disagreement pointed to a need for improved chemical kinetics. A 35-species chemical kinetic mechanism was used to study formaldehyde concentrations in our standard flame configuration under normal gravity conditions with an eye toward understanding the relationship of formaldehyde production and heat release rates. A recent computational effort re-analyzed our standard flames at both normal and zero gravity using a more up-to-date kinetics scheme based on a 66-species C2H4 mechanism. Analysis of other dilution levels is currently underway. To accommodate the constraints imposed by the ISS, the size of the fuel tube and the coflow needed to be changed. Implementing these changes in the computational model requires a multiple-step process to generate converged solutions for the new chemical and physical specifications. This process is now complete for our standard, moderately diluted methane flames. Another area that requires further study in a microgravity environment is at the other end of the fuel dilution spectrum – namely pure fuel, sooty flames. Despite the complexities of modeling soot formation in flames using detailed chemistry, the linkage of soot production to radiation and bulk flame properties is so strong that the coupled treatment of this problem has become a necessity for quantitative modeling of flame structure. Recently, we have made significant progress related to the modeling of soot formation in diffusion flames. Computations agree well with experiments at normal gravity. Microgravity diffusion flames are of direct relevance to the study of soot formation/oxidation in turbulent diffusion flames as these systems are generally non-buoyant due to the large flow velocities. Additionally, microgravity flames have a significantly higher propensity to soot due to increased residence times. Consequently, a study of sooting tendencies at microgravity will provide invaluable information that can be used to improve further our soot formation model. Under normal gravity conditions, we have worked to validate further our soot formation model by

studying the interdependency of soot and NOx formation and to investigate sooting tendencies in time-varying flames.

The soot growth model has now been implemented in our “zero-g” coflow diffusion flame program. Previous soot computations for methane were undertaken with an older version of our methane reaction mechanism. The computations have been updated using “San Diego Mech” by adding new species and pressure-dependent reaction rate coefficients to the program. Those computations are now complete for pure methane under both normal and zero gravity conditions.

Microgravity Experiments

We have been able to benefit from the existence of a closely related set of experiments recently completed on board the ISS. The Smoke Point In Co-Flow Experiment (SPICE) measures the laminar smoke-point properties of non-buoyant jet diffusion flames (i.e., the properties of the largest laminar jet diffusion flames that do not emit soot) for several fuels under different nozzle diameter/co-flow velocity conditions. These studies were designed to improve the understanding of heat transfer mechanisms for different flame conditions, provide insights into soot generation and oxidation processes under microgravity conditions, and devise better fire safety strategies in spacecraft. Dr. David Urban (PI) of NASA Glenn Research Center and Prof. Peter Sunderland (Co-PI) of the University of Maryland generously agreed to share some of their data with us and even allowed us to specify test conditions for further experiments related to our project goals. The similarity of the SPICE flame configuration to our own coflow flames provided an excellent opportunity for us to perform preliminary experiments in a high-quality microgravity environment and to test and improve our analysis strategies in order to obtain the most quantitative data possible from the diagnostics available on board the ISS.

While all of the fuel for the SPICE experiment has been consumed, we were able to define a new subproject entitled “Structure and Liftoff in Combustion Experiments” (SLICE) that will take advantage of the SPICE hardware that is already on orbit. The goal of the SLICE experiment is to investigate the structure of lifting and lifted flames, where the flame detaches from the nozzle and stabilizes at a downstream position. The results will be used to minimize risk and maximize the scientific return of the CLD Flame Experiment. This will enable refinement of the flame diagnostic settings and test matrix for the CLD Flame experiment. It will also allow for a preliminary assessment of the numerical codes’ ability to predict the flame structure under the broader conditions that are only found in microgravity flames.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
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Articles in Peer-reviewed Journals	Dworkin SB, Schaffer AM, Connelly BC, Long MB, Smooke MD, Puccio MA, McAndrew B, Miller JH. "Measurements and calculations of formaldehyde concentrations in a methane/N2/air, non-premixed flame: Implications for heat release rate." Proceedings of the Combustion Institute 2009;32(1): 1311-8. http://dx.doi.org/10.1016/j.proci.2008.05.083 , Jan-2009
Articles in Peer-reviewed Journals	Connelly BC, Long MB, Smooke MD, Hall RJ, Colket MB. "Computational and experimental investigation of the interaction of soot and NO in coflow diffusion flames." Proceedings of the Combustion Institute 2009;32(1):777-84. http://dx.doi.org/10.1016/j.proci.2008.06.182 , Jan-2009
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